

Q1) MCQs or Short Questions:

- a) You are given the sequence of numbers: [11, 3, 16, 1, 9], These numbers are to be inserted one by one into a Max-Heap. Show the state of the trees after each insertion and also after you are done inserting the numbers, show the states after two deletions. Draw the simulation on the opposite page. [3]
- b) What is the time complexity for deleting the minimum value from a Min Heap?
i) $O(1)$ ii) $O(n \log n)$ iii) $O(n)$ iv) $O(\log n)$ [1]
- c) In a Max Heap, the maximum element is located where?
i) Root ii) Any leaf node iii) Leftmost leaf iv) Cannot be determined [1]
- d) Which of the following arrays can represent a Max Heap? [51, 35, 41, 12, 5, 23]
i) [51, 41, 32, 12, 5, 35] ii) [51, 35, 41, 12, 5, 23] iii) [51, 35, 41, 60, 5, 23] iv) [51, 35, 41, 12, 60, 23]

Q2) Write a function called **rangeOutput(root,l,r)** that takes the root of a binary search tree, starting range l and the ending range r; and **prints** the values within that range **in descending order**.

NOTE: Utilizing the BST properties avoid unnecessary traversal. Assume the BTNode class is defined with three instance variables: elem(int), right(BTNode), left(BTNode). **You must use recursion.** You are not allowed to create any other data structure or use static or instance variables. You are allowed to use helper functions if needed.

Given Tree	Output	Explanation
<pre> 50 / \ / \ / \ 20 70 / \ / \ 10 30 60 80 / \ 75 90</pre> Input: rangeOutput(root,15,78)	75,70,60,50,30,20	We are given the root of a BST, the starting range 15 and ending range 78. We printed only the values within the given range in descending order starting from 75 to 20.